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Does e-government integration contribute to the quality and equality of local public services? Empirical evidence from China

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Integrated e-government at different scales plays an increasingly important role in local public services worldwide. This study uses panel data from 31 provincial e-governments in China from 2015 to 2020 to investigate the effects of the intraprovincial e-government integration on the provincial quality and interprovincial equality of public services. The results show that integrated e-government effectively contributes to the quality and equality of public services. Mechanistic tests reveal that integrated e-government promotes service quality by improving administrative efficiency and supporting technological innovation. Under central government support, integrated e-government also improves the interprovincial equality of public services by reducing administrative inefficiency from decentralization and promoting technology diffusion. Moreover, e-government integration is a stronger driver of service quality and equality for provinces with low fiscal transparency. These findings provide theoretical and practical insights for promoting public service quality and equality through collaborative e-government.

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Introduction

Public service quality and equality are among the most important policies that aim to ensure compliance with the directive to "leave no one behind (LNOB)" (Stokan et al. 2022). With the digital transformation of governments worldwide, assessing whether and how e-government integration affects the quality and equality of local public services is critical for exploring the inequality theme underlying the UN Sustainable Development Goals (SDGs) (UN, 2016, 2023).

Public services involve basic human demand for survival and development, including education, health care, labour and social benefits, which are important parts of the public welfare system supplied by the local government (Cepiku and Mastrodascio, 2021). The quality of local public services refers to the coverage and degree of citizens whose rights are guaranteed (Dai et al. 2022). Furthermore, considering that public service is a kind of redistribution, its equality will narrow the income gap and promote social and economic development both sustainably and equitably (Ullah et al. 2024; Clements et al. 2024).

The impacts of e-government on public services have attracted increased attention from professional and academic communities. In terms of quality, scholars have shown that digital support services not only enable public infrastructures with better energy, computing competency and connectivity but also transfer limited resources effectively (Cai and Zhang, 2023; Yigitcanlar et al. 2023). Compared with changes in service forms and channels, organizational changes are undervalued (Luna et al. 2024). In terms of equality, most previous studies have discussed e-government and equality across groups (Larsson, 2021), while the relationship between information technology and interregional equality remains uncertain. A positive opinion holds that e-government improves the urban and rural equality of public services (Liu and Yuan, 2024); in contrast, a negative opinion considers the presence of a positive correlation between the digital divide and development disparities (Alp and Baycan, 2024).

Although integrated e-government effects have been discussed in recent years, such effects have not been connected with common factors related to the quality and equality of public services. Hence, the relationships between e-government integration and public services and the related mechanisms have been understudied. Common factors related to quality include government operations and basic conditions, such as economic development, population size, and the natural environment (Li et al. 2017; Yang et al. 2019). For regional equality, common factors refer to decentralization, autonomy, and differences in common development conditions (Niu, 2013; Bucci et al. 2024). E-government integration may not only contribute to government operation (Margariti et al. 2022) but also overcome disparities in the willingness and ability of public services caused by decentralization (Yan and Lyu, 2023).

To fully explore the effects of integrated e-government on the quality and equality of local public services, a panel analysis of 31 provinces in China is performed in this study. The challenge of achieving the quality and equality of local public services is significant in China, as it is the largest developing country in the world and its government has five levels and has vigorously implemented decentralized practices in recent decades, including the central, provincial, municipal, county and town levels (Li et al. 2023). Due to the country's subnational variation and a large population, the China State Council (the Cabinet) has highlighted the National Integrated Government Service Platform for integrating online and offline IT infrastructures and promoting data sharing and business collaboration across regions, functions, and levels (Mao and Zhu, 2023). Furthermore, because strong links and political pressure from central to local governments in the Chinese unitary system help this integration more easily than do federated states (Henman et al. 2021), all 31 intraprovincial e-governments were integrated from the provincial level to the town level by the end of 2021.

The remainder of the paper is organized as follows. In Section 2, the mechanism analysis and research hypotheses are presented after the literature is reviewed. In Section 3, the methodology and data of the study are described. In Section 4, the main empirical results are reported. Finally, in Section 5, the conclusion and targeted implications are presented.

Theoretical framework

Literature review

Definition and measurement of public service quality and equality. Public services consist of the essential services produced by the government or public institutions that aim to safeguard fundamental human rights, involving basic needs such as education, labour, and health care (Cepiku and Mastrodascio, 2021). Thus, everyone should obtain essentially the same minimum public service (Kuehn, 2021). To reflect the outcome of public service provision, the quality of regional public services usually utilizes the composite index of per capita indicators in several areas, such as education, labour, health care (Dai et al. 2022). To capture its substantial change, in this study, the composite index is divided by the provincial average as the provincial quality (Li et al. 2017), and the composite index is used in a robustness test.

Although equality is an internal attribute of public service, there is still inequality across individuals, groups, regions, etc. (Stokan et al. 2022; Elgin et al. 2023). To measure interregional inequality in public services, there are two main comparison approaches. From an overall perspective, some studies have used the Theil index, Gini coefficient and kernel density estimation to describe spatial distribution and concentration (Li et al. 2024). However, the benchmarking method has also been recognized for better highlighting the gap between a single region and other regions, such as when the desired degree is used as a benchmark (Guo et al. 2023). Therefore, in this study, interprovincial inequality is measured by the relative gap between the service quality of a given province and the highest quality among provinces (Yuan et al. 2024).

Common influence on provincial public service quality. Provincial public service quality usually involves government operations and basic conditions, such as economic development, population density, and the natural environment (Li et al. 2017; Yang et al. 2019). The provincial quality of public services is the result of collaboration across multiple functions and hierarchies, characterized by asymmetric information, inadequate accountability, painful coordination and implementation inertia (Ma, 2014; Chen et al. 2019).

Some studies have found solutions to the abovementioned problems in the fiscal and technical dimensions. Fiscal transparency could improve public service quality by optimizing the fiscal expenditure structure and improving the efficiency of public service funds (Yang, 2021). ICTs enable a transparent public service delivery system and guarantee reliability in terms of the quality and timeline of service fulfillment (Venkatesh et al. 2016).

Common influence on interprovincial public service inequality. In addition to disparities in common development conditions, interprovincial inequality is derived mainly from fiscal and administrative decentralization, leading to differences in the motivations and abilities of local public services (Niu, 2013; Bucci et al. 2024). Initially, the central government transfers control

over public resources to lower levels of government, which provide better services according to local cost and demand than does the central government (Costa-Font, 2010). However, whether each local government provides relatively equal services depends on institution-specific designs, such as financial and human resources, regulations, and the responsiveness of local authorities (Cepiku and Mastrodascio, 2021).

In reality, local public service systems may be limited by autonomy capacity, elite capture and corruption, thereby increasing regional disparities (Zarychta et al. 2024). Antonymy refers to principal–agent problems and uncertain public services, especially when GDP-oriented promotion and polarized fiscal self-sufficiency push officials to invest more in economic infrastructures than in long-term public services (Chen et al. 2018; Li et al. 2023). Interprovincial transfer payments that are guided by the central government constitute a fiscal system for reallocating resources within provinces with different financial capabilities; however, supervision is needed because of the problem of embezzlement (Zhang and Tang, 2013).

Development of integrated and decentralized e-government. The different carriers and forms of integrated and decentralized e-government in the past have shifted from a front-door interface to one that is more backstage in nature (Mu and Wang, 2024). During the popularization of computers in governments in the 1980s, initial e-government efforts focused on an online presence, with the goal of organizing scattered electronic documents (Layne and Lee 2001). During the development of government online portals in the 1990s and 2000s, centralized national web portals adopted an information repository design, whereas the decentralized type defaulted to an information referral design (Henman et al. 2021). The centralized structure contains substantial information, needs coordination and is developed mostly in unitary and city states (i.e., the UK and Singapore) (Clarke, 2020). With the recent development of emerging technologies, blockchain has become the prevailing solution for decentralized public information, and self-regulated digital networks have increased levels of trust and security in public services (Kassen, 2023; Clifton et al. 2023).

Currently, the integrated model of e-government exists worldwide at different scales and degrees, aiming at better public services, although the political environment may limit national integration (Alexopoulou, 2024). Whatever the administrative system, a service-oriented view requires an interorganizational information system project to integrate data (Chen and Lee, 2018). From an institutional perspective, specialized e-government units have emerged globally at both the national and subnational levels (i.e., the US and Canada) and are dedicated to broader integration (Clarke, 2020). From a technical perspective, the general commonalities of "government as a platform" include platform infrastructures with shared IT components and centralized core functionalities (i.e., the UK and Estonia) (Kuhn et al. 2023). From a democratic perspective, the open architectures of integrated e-government allow a broader range of actors, thereby expanding civil society participation and making datasets and APIs available for third-party tech companies to build new public services (Aguerre and Bonina, 2024; Jiang and Fan, 2024).

Potential effects of integrated e-government on public service quality. Integrated e-government, which is characterized by integrated information resources and processes and the interoperation of diverse systems, supports a shared, synergistic and holistic model of government operation (Margariti et al. 2022).

The larger combination of government entities across institutional, functional and geographic boundaries makes it possible for a series of organizations to advance public service quality both widely and effectively (Schooley and Horan, 2007; Abbas et al. 2024). The first direct benefit of integration is technical improvements related to data management, which promote better standardization and technical application in public services within a region (Scholl et al. 2012). The second benefit is the organizational benefit related to strengthening organizational capacity, thereby connecting decentralized authority and functions at all levels of government through the network (Gil-Garcia and Sayogo, 2016). Finally, political benefits provide more comprehensive public information, accountability, and integrated planning and services (Zeng et al. 2023).

Conversely, fragmented e-government is in its initial stage when reality is mapped onto the virtual world, representing an enhanced version of traditional government (Layne and Lee, 2001; Yang and Wu, 2014). This model can still lead to problems such as inadequate internal control, the amplification of discretion, and a shortage of public services in some local governments (Hoffman and Karpiuk, 2022).

Potential effects of integrated e-government on public service equality. The dual effect of e-government on public service equality has attracted an increasing amount of attention. On the one hand, e-government has the potential to mitigate geographic barriers and bureaucratic impediments to equal access (Alexopoulou, 2024). E-government reduces the utilization of personal connections with government officials for resource allocation (Fan and Ma, 2024). At the regional level, e-government improves the urban and rural equality of public services by improving the efficiency of supply and transfer payments (Liu and Yuan, 2024).

On the other hand, e-government raises concerns about the digital divide across groups or regions. Digital defaults in e-government create a new norm of exclusion or marginalization because digital administrative burdens or data mistakes erect barriers to public service for some citizens (Al-Muwil et al. 2019; Larsson, 2021). The number of e-service use gaps generally decrease in the long term (Sung and Lee, 2024). At the regional level, there is a positive correlation between the digital divide and development disparities (Alp and Baycan, 2024).

Mechanism analysis and research hypotheses

Direct effects of e-government integration on the provincial quality and interprovincial equality of public services. Integrated e-government facilitates rational decision-making and scientific resource utilization for public service quality. Prior to the policy-making stage, e-government covers various types, groups, areas and departments of public services, gathering comprehensive information for subsequent analysis (Wouters et al. 2023). By identifying patterns, trends and issues in public services, insufficient education resources or the low coverage of medical resources in certain areas can be identified (Curtis, 2019). As public service providers gain better insights into citizens' needs through more complete data, they can learn, adjust and respond to these needs collectively in a timely way (Larsen and Følstad, 2024). Online connectivity helps clarify the division of labour, encourages participating departments to be accountable to one another and the citizens they serve, and effectively transforms public resources into public services (Chen et al. 2019).

An integrated e-government could overcome the negative effects of some interprovincial factors that cause an interprovincial public service gap, such as decentralization and selective policy implementation. For disparities in intentions, e-government integration across all administrative hierarchies allows every province to be directly supervised by the higher-

ranking authority, thereby ensuring enough public service input and mitigating economic intergovernmental competition (Ma and Zheng, 2019; Silal et al. 2023). With respect to disparities in capabilities, shared infrastructures and procurement assistance can potentially solve the shortage of financing and skills, thereby creating lean and consistent governments (Cordella and Paletti, 2019; Kuhn et al. 2023). An integrated e-government also makes it convenient for provinces to communicate, collaborate, and even promote some public service projects together, consequently narrowing the interprovincial gap (Luna-Reyes et al. 2007; Ramírez de la Cruz et al., (2020)). As more provincial public services are recorded and aggregated online, the central government will be better able to allocate interprovincial resources (Yan and Lyu, 2023).

Hypothesis 1 (H1): The integration of e-government promotes the provincial quality and interprovincial equality of public services.

Mediating mechanisms linking e-government integration to provincial quality and interprovincial equality. The integration of e-government connects silos and streamline procedures, facilitating the smooth operation of public services. Initially, the speed and ease of use of online tools relieves government employees and saves more human resources and administrative costs related to public services (Kuhlmann and Heuberger, 2023). By improving transparency and removing red tape, the delivery of public services also becomes more friendly for citizens, resulting in a more effective production process (Arayankalam et al. 2021). The underlying information accessibility and connectedness also support the flexibility and performance of large-scale governance structures, such as avoiding investment duplication and resource waste in public services (Fan et al. 2019).

Provincially integrated e-government reduces administrative inefficiency caused by decentralization, enhancing the efficiency of interprovincial public issues and coordinating public services among provinces. The computerization of provincial government operations minimizes the opportunities for policy implementation to deviate from national standards, enables provinces to strictly adhere to their budgets and reduces loss from discretion (Prasad and Shivarajan, 2015; Fan and Ma, 2024). Less vicious interprovincial governmental competition brings responsive provincial governments closer to public demands, resulting in balanced public services among provinces (Karkin and Janssen, 2020). Under the requirements of the central government, provincial e-governments support cross-provincial government services (Liu et al. 2022). This type of data sharing helps some provinces compare themselves with others and fill the public service gap after data analysis (Boudreau and Bernier, 2017; Alexopoulou, 2024).

Hypothesis 2 (H2): The integration of e-government promotes the provincial quality and interprovincial equality of public services by improving administrative efficiency.

Integrated e-government provides ingredients, unified infrastructures and data flows for technology innovation in public service quality. Multilevel and multifunctional integration under political pressure unblocks vast amounts of data, driving technology innovation in the fields of data processing, predictive analytics and decision support for public services (Ma et al. 2020). During this process, provinces provide a wider testing ground for emerging technologies, where inclusive regulation on the practical application of technologies in public services benefits the iterative upgrading of technologies according to real demand (Eppel and Lips, 2016). Additionally, the large-scale construction of e-government attracts private technology suppliers, increasing the efficiency of technology innovation in public areas (Feng and Nie, 2024).

Interprovincial technology diffusion based on integrated e-government can promote technology innovation, thereby narrowing the interprovincial gap in public services. A fragmented IT hierarchy increases the costs and conflicts of migration, hindering technology updating (van Loon and Toshkov, 2015; Uster, 2024). Conversely, e-government integration establishes a standardized framework and technical interoperability, thereby reducing the barriers to technology adoption and diffusion across provinces (Kamal et al. 2015; Zhao et al. 2017). On this basis, provinces reproduce and reuse rapidly and make technological adjustments per local public service, translating integrated data and knowledge into innovativeness and avoiding public service polarization through a provincial digital divide (Epstein, 2022).

Hypothesis 3 (H3): The integration of e-government promotes the provincial quality and interprovincial equality of public services by supporting technology innovation and its diffusion and localization.

Moderating effects of fiscal transparency. Considering the current literature on the contribution of e-government to government transparency (Gil-Garcia et al. 2018), in this study, it is proposed that the impact of e-government integration may be moderated by fiscal transparency. Fiscal transparency requires the greatest amount of disclosure possible of relevant fiscal information, resulting in the government functioning with lower levels of information costs and policy uncertainty (Guillamón et al. 2011).

In terms of provincial quality, e-government may compensate for low levels of fiscal transparency and contribute to public performance (Zhu and Yu, 2024). Similarly, the quality of e-government may be more weighted for FDI decisions if the host country is corrupt because enterprises may seek to control risks through exploiting e-government (Kim and An, 2022).

For interprovincial equality, fiscal transparency is a key solution for Chinese local financial expenditure preferences (Yan and Lyu, 2023), which is an important factor for interprovincial inequality. In provinces with lower levels of transparency, problems such as the misappropriation of fiscal funds for public service are more severe; thus, the potential for technology to solve such problems could be greater (Martins et al. 2023) and have a more prominent effect on interprovincial equality. Similarly, e-government plays an active role in supervising transfer funds for the regional equality of public services (Liu and Yuan, 2024).

Hypothesis 4 (H4): In provinces with lower levels of fiscal transparency, the potential for e-government integration to contribute to the provincial quality and interprovincial equality of public services could be greater.

The path of e-government integration in promoting the quality and equality of local public services is summarized in Fig. 1.

Methodology and data

Empirical models. To test Hypothesis 1, a two-way effect model is applied in this study in view of the interaction between time and individuals. To further verify Hypotheses 2 and 3, the mediating effect method designed by Wen et al. (2004) is borrowed to explore mechanisms. In addition, to verify Hypothesis 4, the interaction term between e-government integration and the moderating variable is introduced.

The specific empirical models are presented in Section A1 of Appendix A.

Key variables

Dependent variable. In accordance with current studies and practices (Narbón-Perpiñá et al. 2020; Shi et al. 2023; Cheng and Liu, 2024), a comprehensive evaluation system is constructed that

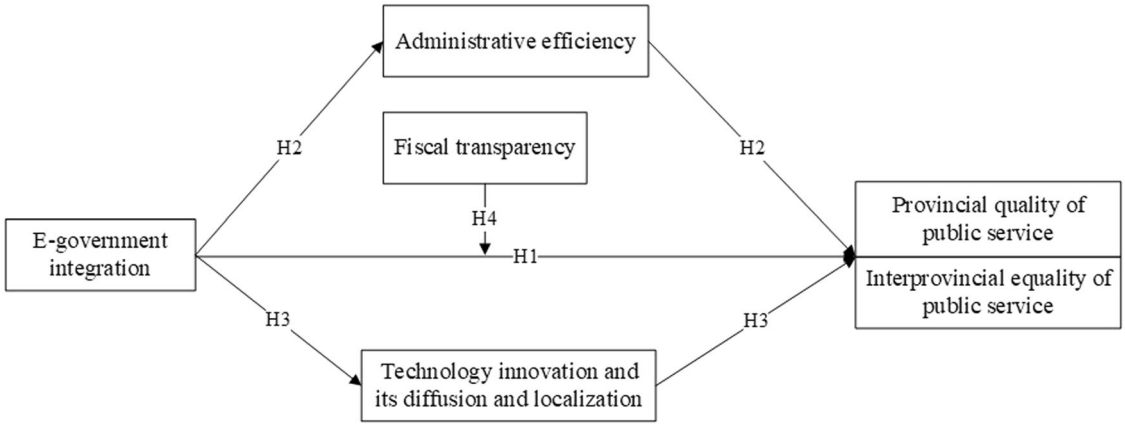

Fig. 1 Research hypotheses diagram. H1, The direct effects of e-government integration on both provincial quality and interprovincial equality of public service. H2, The mediating effects through administrative efficiency. H3, The mediating effects through technology innovation and its diffusion and localization. H4, The moderating role of fiscal transparency on the relationships between e-government integration and public service quality and equality.

Table 1 Evaluation index system for the quality of provincial public services.

Dimension	Indicator	Weight (%)	Efficacy
Education	Number of general primary and secondary schools per 10,000 people	3.84	+
	Pupil–teacher ratio in general primary schools	3.31	-
	Pupil–teacher ratio in general secondary schools	1.95	-
	Number of students in primary and secondary schools per 10,000 people	2.58	+
Labour	Number of vocational guidance per registered job seekers	5.44	+
	Number of entrepreneurship services per registered job seekers	7.75	+
	Employment rate of vocational training institutions	1.01	+
	Settlement rate of labour disputes	1.48	+
Social security	Participation rate in basic pension insurance	1.29	+
	Participation rate in basic medical insurance	2.86	+
	Participation rate in employment injury insurance	7.70	+
	Participation rate in unemployment insurance	6.56	+
	Participation rate in maternity insurance	7.96	+
Social services	Ratio of the average subsistence allowance and per capita consumption expenditure of urban residents	2.97	+
	Ratio of the average subsistence allowance and per capita consumption expenditure of rural residents	2.45	+
	Number of nursing beds per 1,000 elderly people	1.85	+
	Number of social workers per 10,000 people	11.75	+
Health care	Per capita number of outpatient visits	3.56	+
	Daily number of medical services per doctor	3.14	+
	Number of primary healthcare institutions per 10,000 people	3.14	+
	Utilization rate of beds in community health service centres	2.32	+
Culture and sports	Annual circulation of public libraries per 10,000 people	5.28	+
	Participants in cultural activities organized by mass cultural institutions per 10,000 people	5.21	+
	Coverage of cable broadcast television	4.60	+

covers the dimensions of education, labour, social security, social services, health care, and culture and sports when the entropy method is used for the composite index according to Table 1.

To describe its substantial change, herein, the composite index is divided by the provincial average as the provincial quality (Li et al. 2017), and the composite index is used in a robustness test. The equation is as follows:

$$\mathrm{QPS}_{it} = \frac{Composite\ index_{it}}{ave(Composite\ index_{it})} \tag{A2.5}$$

Therefore, in this study, the interprovincial inequality is measured by the relative gap between the service quality of a given province and the highest quality among provinces (Yuan et al. 2024). The equation is as follows:

$$EPS_{it} = \frac{\max\left(Composite\ index_{it}\right) - Composite\ index_{it}}{Composite\ index_{it}} \tag{A2.6}$$

The specific calculation procedures are presented in Section A2 of Appendix A.

Independent variables. E-government integration refers to an organizational arrangement for multiorganization collaboration to create holistic impacts of reusing technology resources and sharing information (Luna-Reyes et al. 2007; Al-Balushi et al. 2016). Considering the representatives and validities, the intra-provincial e-government integration index published by the China National School of Administration is selected as the independent variable in this study. The index comprehensively evaluates the vertical and horizontal integration of e-government,

Table 2 The impact of e-government integration on provincial quality and interprovincial inequality of public services.

Variables	Provincial quality		Interprovincial inequality	
	(1)	(2)	(3)	(4)
lnINTEG	0.287***	0.280***	-0.546***	-0.510***
	(3.262)	(3.515)	(-3.079)	(-3.034)
Economic development		0.064***		-0.133***
		(3.780)		(-3.713)
Urbanization rate		2.531***		-5.182***
		(3.223)		(-3.124)
Fiscal decentralization		1.533**		-2.349
		(1.985)		(-1.440)
Transfer payment		0.601***		-0.772*
		(3.232)		(-1.966)
Population density		-20.569***		22.199
		(-2.727)		(1.394)
Constants	-0.243	-1.470**	3.624***	6.708***
	(-0.637)	(-2.373)	(4.719)	(5.127)
Fixed time and province	Yes	Yes	Yes	Yes
N	186	186	186	186
R^2	0.067	0.273	0.516	0.586

Note: "*", "**", and "***" denote significance at the statistical levels of 10%, 5%, and 1%, respectively.

commissioned by the E-Government Office of the China State Council.

Mediating variables. Administrative efficiency is measured as "1 minus the ratio of general public service spending to GDP in each province" when general public service spending is used mainly to ensure the normal operation of government agencies and support them in performing functions (Srithongrung, 2018).

Technology innovation and its diffusion and localization are represented by the "per capita number of patent grants"; i.e., when there are more patent grants, the technology innovation output of regions is stronger (Xiong et al. 2022).

Moderating variables. The level of provincial transparency is determined by examining municipality-level fiscal transparency information obtained from the Annual Fiscal Transparency Report of Chinese Municipality-level Governments produced by Tsinghua University (Sun and Andrews, 2020). The specific calculation procedures used are presented in Section A3 of Appendix A.

Control variables. Considering that many other factors may affect the quality and equality of local public services(Narbón-Perpiñá and De Witte, 2018; Yuan et al. 2024), in this study, the control variables of economic development, the urbanization rate, fiscal decentralization, transfer payments and population density are introduced. Economic development is measured by GDP per capita. The urbanization rate is denoted as the ratio of the permanent urban population to the total population. Fiscal decentralization is denoted as the ratio of a provincial government's expenditure to the central government's expenditure, representing the relative autonomy to manage local affairs. Transfer payments are denoted as the ratio of transfer payments from the central government to provincial fiscal expenditures, which indicates the help for insufficient local public services. The population density is denoted as the ratio of the provincial population to the administrative area.

Chinese context and study sample. According to the 2022 E-government Survey conducted by the United Nations, China's ranking in the E-Government Development Index (EDGI) rose from 78th in 2012 to 43rd in 2022, marking one of the highest increases worldwide. Since the 1990s, China has advanced the concept e-government and promoted the "Gold Card, Golden Bridge and Golden Pass" initiative, marking the beginning of information construction (Yan and Lyu, 2023). Circa 2000, the Government Internet Project and Office Automatic (OA) pushed the e-government concept into a stage of large-scale construction. In the past decade, Chinese e-government has gradually moved from technological application to integration. In particular, a national integrated government with the National Integrated Government Service Platform serving as its main hub connects 31 provincial e-governments, which represent the first level of local government in mainland China. These 31 intraprovincial e-governments also integrate infrastructures and part of the data at the municipal, county and town levels (Mao and Zhu, 2023; Fan et al. 2023).

E-government integration is more likely to emerge in institutional ecologies with centralized administrative authorities (Henman et al. 2021). Additionally, complex relationships may exist between e-government integration and local public services in China, where local governance involves 34 provinces, 333 municipalities, 2843 counties, and 38602 towns, ultimately covering approximately 1.4 billion people and 9.6 million square kilometres. Consequently, China contributes to the appropriate selection of an empirical sample scale and provides an ideal setting for verifying the impact of e-government integration on the quality and equality of local public services.

This study covers 31 provinces, with the exceptions of Hong Kong, Macao and Taiwan, because data are unavailable from those regions. Hence, the data sample of this study includes information drawn from 31 administrative regions from 2015 to 2020, which ultimately produces a balanced panel with 186 observations. The starting year is 2015 because the China National School of Administration began releasing the official evaluation of Chinese e-government integration to the public in 2015. The basic data are drawn from the Statistical Yearbook of China, China Education/Labour/Civil Affairs/Health/Culture and Related Industries Statistical Yearbook, etc. The descriptive statistics of the raw data are presented in Section A4 of Appendix A.

Empirical results

Baseline regression analysis results. Compared with the pooled model and random effects model, the fixed effects model is more

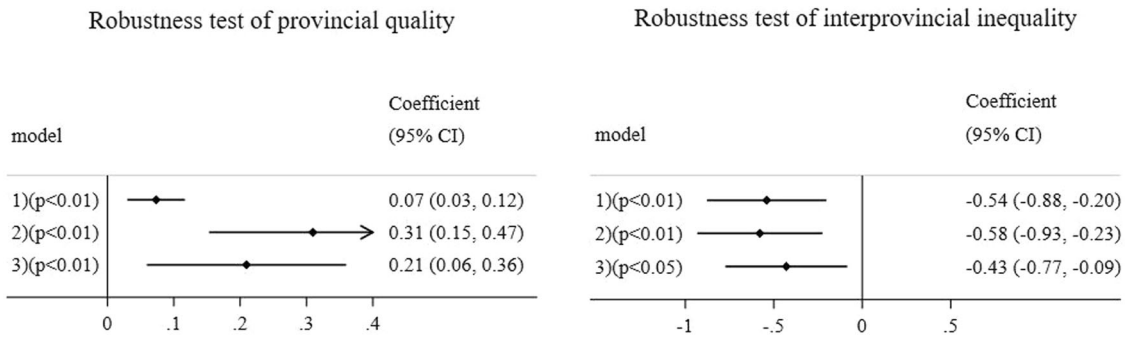

Fig. 2 Robustness test results.

appropriate for conducting regression analysis in this study. The results of the F test, LM test, Hausman test and Sargan-Hansen test support the use of the fixed effects model for both provincial quality and interprovincial equality.

The results of the regression analysis are shown in Table 2. Introducing control variables based on Column (1), the regression results in Column (2) still indicate that there is a significant positive effect of e-government integration on provincial public service quality ($\alpha_1 = 0.280$, $p < 0.01$). From Columns (2) to (4), the regression results indicate that e-government integration negatively and significantly affects interprovincial public service inequality ($\alpha_1' = -0.510$, $p < 0.01$). Thus, Hypothesis 1 is verified.

Robustness test. In this study, the robustness of both the model and results is tested by 1) changing the measurement, 2) excluding special provinces and 3) excluding special years. First, the quality is adjusted to the composite index obtained via the entropy method, and the inequality is synthesized based on the composite index obtained via the entropy method combined with the TOPSIS method. Four municipalities are excluded because of the particularities of their policy environment and economic development, which may lead to great differences from those of other provincial-level governments. Given the impact of the COVID-19 epidemic on the public service supply, post-2019 samples are also dropped. As shown in Fig. 2, the estimated coefficients, effect directions and significance levels of e-government integration are consistent with the baseline regression results, again validating the contribution of e-government integration to the provincial quality and interprovincial equality of public services.

Endogeneity test. Regarding potential endogeneity problems, herein, an instrumental variable that is related to e-government integration but uncorrelated with the quality and equality of basic public services is needed. Thus, the per capita throughput of postal and telecommunications in 1984 is selected as the instrumental variable. From the perspective of correlation need, a larger volume of postal and telecommunications business can reflect the greater construction of postal and telecommunications infrastructures in local history, which may provide important support and a basis for the follow-up development of e-government (Zhu and Yu, 2024). In contrast, the impact of postal and telecommunications business a few decades ago on basic public services has gradually decreased, thereby meeting the exclusivity requirement. Because the abovementioned instrumental variable consists of cross-sectional data that remain constant over time, it cannot be directly applied in panel data analysis. Thus, referring to previous studies (Nunn and Qian, 2014; Yi et al. 2022), a time series variable is introduced and the interactive term of the per capita throughput of post and telecommunications in 1984 with the R&D expenditure of industrial enterprises above a designated size in that term is constructed, which is treated as the instrumental variable of e-government integration.

Table 3 Endogeneity test results of provincial quality and interprovincial inequality of public services.

Variables	Provincial quality		Interprovincial inequality	
	1st	2nd	1st	2nd
ln(POST*RD)	0.170*** (4.263)		0.170*** (4.263)	
lnINTEG		0.994*** (3.474)		-1.803*** (-3.137)
Control variables	Yes	Yes	Yes	Yes
Fixed time and province	Yes	Yes	Yes	Yes
Observations	186	186	186	186

Note: "*", "**", and "***" denote significance at the statistical levels of 10%, 5%, and 1%, respectively.

The results of the two-stage least squares (2SLS) estimation are shown in Table 3. The altered coefficients indicate that e-government integration still has significant effects on the provincial quality and interprovincial inequality of public services even after endogeneity is eliminated. Moreover, the Anderson canonical correlation LM tests and Cragg–Donald Wald F tests reject the original hypotheses of underidentification problems and weak instrumental variables, respectively. Thus, the results of the baseline regression are again verified to be robust.

Mediating effects results. The value of the coefficient in Column (1) is 0.014, with a significance level of 5%, reflecting the contribution of integration to government operation. The regression coefficient of administrative efficiency in Column (2) is 5.392, with a significance level of 1%, indicating its positive effect on public service quality. The coefficient of integration is 0.205, and its absolute value is less than that in the baseline regression. This means that administrative efficiency is a partial mediator of the positive nexus between e-government integration and provincial quality. Additionally, Column (3) indicates that administrative efficiency is a partial mediator between integrated e-government and interprovincial inequality. Columns (4), (5) and (6) verify that technology innovation and its diffusion and localization constitute another partial mediator. In short, Hypotheses 2 and 3 are verified in Table 4.

Moderating effects results. Table 5 presents the moderating effects of fiscal transparency. In Column (1), the coefficient of the interaction term is −0.396, which is significant at the 1% level,

Table 4 Mediating mechanism results.

Variables	(1) Administrative efficiency	(2) Provincial quality	(3) Interprovincial inequality	(4) Technology innovation	(5) Provincial quality	(6) Interprovincial inequality
lnINTEG	0.014**	0.205***	-0.384**	0.540***	0.216***	-0.389**
	(2.138)	(2.814)	(-2.393)	(2.694)	(2.765)	(-2.335)
Administrative efficiency		5.392***	-9.108***			
		(5.826)	(-4.472)			
Technology innovation					0.118***	-0.224***
					(3.711)	(-3.318)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Fixed time and province	Yes	Yes	Yes	Yes	Yes	Yes
Observations	186	186	186	186	186	186

Note: "*", "**", and "***" indicate significance at the statistical levels of 10%, 5%, and 1%, respectively.

Table 5 Moderating effects results.

Variables	Fiscal transparency	
	(1) Provincial quality	(2) Interprovincial inequality
lnINTEG	1.754***	-2.560**
	(3.765)	(-2.538)
lnTransparency	1.733***	-2.429**
	(3.308)	(-2.142)
lnINTEG*	-0.396***	0.551**
lnTransparency	(-3.211)	(2.062)
Control variables	Yes	Yes
Fixed time and province	Yes	Yes
Observations	186	186

Note: "*", "**", and "***" denote significance at the statistical levels of 10%, 5%, and 1%, respectively.

indicating that fiscal transparency plays a weakening role in e-government integration, positively affecting provincial quality. This suggests that the positive effect of e-government integration is mitigated in provinces with higher levels of fiscal transparency. Similarly, Column (2) indicates that fiscal transparency also plays a weakening role in e-government integration, negatively affecting interprovincial inequality. In brief, Hypothesis 4 is verified.

Conclusion and discussion

Main findings. To examine the impacts of integrated e-government on local public service quality and equality, this study conducts an empirical investigation on the basis of panel data drawn from 31 Chinese provinces from 2015 to 2020. The results demonstrate several key findings. First, the direct impacts of e-government integration on both the provincial quality and the interprovincial equality of public services are positive. These conclusions are supported by a series of robustness tests. Furthermore, administrative efficiency and technology innovation are intrinsic mechanisms through which integrated e-government addresses the provincial quality of public services. Moreover, under central government support, integrated e-government also improves the interprovincial equality of public services by reducing administrative inefficiency due to decentralization and promoting technology diffusion and localization. The heterogeneity analysis reveals that for provinces with lower levels of fiscal transparency, e-government integration has a stronger driving effect on public service quality and equality.

Theoretical contributions. The marginal contributions for digitalization and the SDGs are as follows.

First, in terms of research on public service quality, most studies have focused only on technology applications in public sector internal operations and service delivery and ignored their substantive impact on public service outcomes and related mechanisms. This study proves that integrated e-government has positive effects on public service quality, which is in line with the results of previous studies on the smart city effect (Cai and Zhang, 2023). This study reveals that this effect occurs by improving levels of administrative efficiency and technology innovation, further confirming certain positive views on online offices and data-driven public services (Camarena and Fusi, 2022; Alexopoulou, 2024).

Second, in terms of research on the interregional equality of public service, most previous studies have been concerned with the problem of local autonomy resulting from decentralization, which is the main cause of interprovincial inequality (Niu, 2013; Bucci et al. 2024). This study verifies that integrated e-government indeed promotes interprovincial equality by reducing inefficiency in local autonomy, providing a new solution that acts as a type of regulation mentioned in previous literature (Cepiku and Mastrodascio, 2021).

Third, in terms of the uncertain relationship between digitalization and interregional equality, some studies have explored how e-government contributes to urban–rural equality by improving supply efficiency (Liu and Yuan, 2024), whereas other studies have reported a positive correlation between the digital divide and regional development disparities (Alp and Baycan, 2024). Considering that interprovincial technology diffusion is guided by the central government (Yang et al. 2024), this study finds that integrated e-government narrows public service gaps through technological learning and localization.

Fourth, in terms of the research on integrated e-government, most studies have discussed the advantages of the integrated e-government model from organizational dimensions using qualitative methods (Ingrams et al. 2018; Zhang and Kimathi, 2022). This study follows past research on sharing and standardization characteristics, further verifying the advantages of transparency, efficiency and participation, which benefit public services. The Chinese case makes quantitative tests at the macro level available, further supporting some general conclusions.

Practical implications. For public service quality, local governments could advance technology applications in service delivery to multiple-source data fusion in policy design. Projects guided by

a single department, such as smart education, can easily duplicate investments and waste resources. Although public services are handled by different departments for education, health care, culture, etc., a comprehensive investigation of demand and supply still requires integrated data across levels and departments. On this basis, local governments would be able to identify vulnerable groups or areas, ageing populations, unemployment, etc., and improve resource allocation efficiency.

For regional equality in public service, the central government should play a role in coordinating subnational governments through integrated e-government. From an international perspective, central coordination and top-down support constitute a core shift in digital-era government (Alexopoulou, 2024); however, there is insufficient interoperability and integration in some highly decentralized federal systems (Kuhlmann and Heuberger, 2023). Therefore, the central government is encouraged to guide interprovincial collaboration and technology diffusion. In terms of policy, standards for cross-provincial services are necessary. In organizational management, establishing communication and incentives for personnel training and technical exchange is a preparation for basic systems. In this way, integrated e-government not only overcomes the negative effect of decentralization on interprovincial inequality but also avoids the potential impact of the digital divide.

For e-government development, the advantages of the integration model could be exploited in any interorganizational information project. Regardless of the political environment, the integrated model of e-government has existed globally at different scales and degrees over past ages (Alexopoulou, 2024). Inspired by the findings of this case, higher-ranking authorities and shared values and rules would advance complex collaborative e-governance, which also aligns with current e-government exploration of soft political and hard technical aspects, such as the emergence of e-government units and platforms in Australia and Canada.

Limitations and future research. In addition to the above-mentioned findings and implications, this study has several limitations and directions for improvement. First, owing to the limitation of the use of subjective data from citizens, this study considers only the quality and equality of public services from suppliers, not receivers. With the expansion of various survey data archives, it is possible to complement the current study with data on citizen satisfaction levels and perceptions of equality. Second, this study explores the impact of e-government integration on public services in only Chinese provinces; this impact should also be explored in other developing countries. Third, the mediators used in this study are all partial mediators. Therefore, some unstudied mediators must be considered in the future to advance the related literature.

Data Availability

The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

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Zijun Mao: Acquired funding, reviewed the draft and supervised process. Yuqian Zhu:Clarified the conceptual logic, wrote the paper and further modified the content.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval was not required as the study did not involve human participants.

Informed consent

Informed consent was not required as the study did not involve human participants.

Additional information

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